



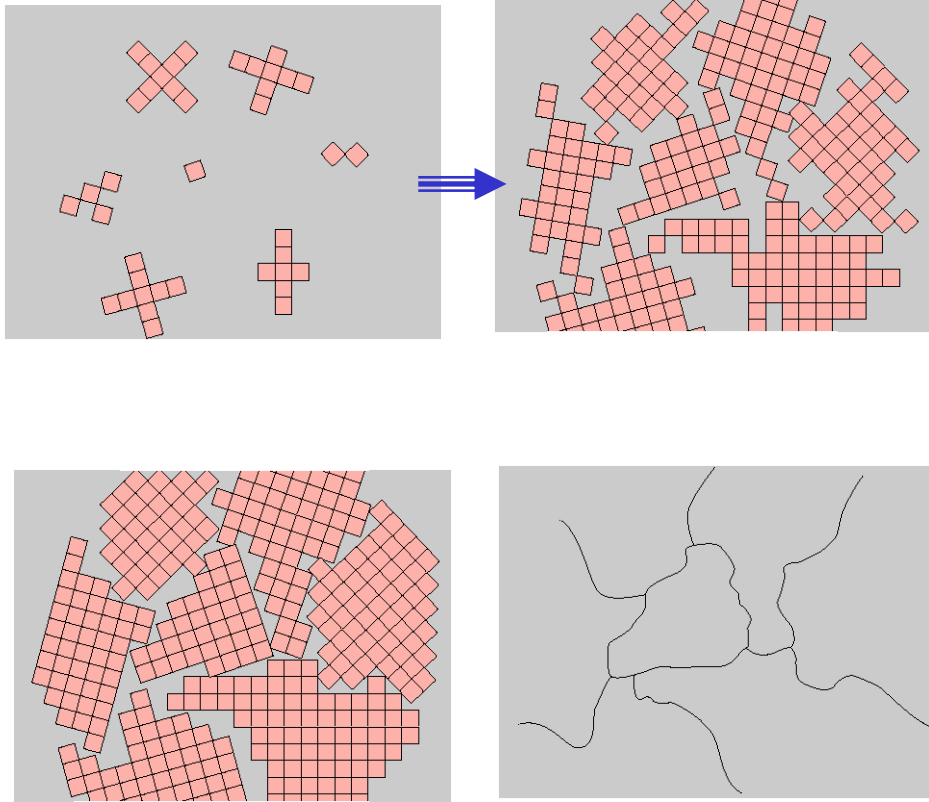
Outline:

- Crystallization
- Grain boundaries
- Grain size determination
- Stacking faults
- Types of microscopes
 - *LOM*
 - *SEM*
 - *TEM*
 - *SPM*



Polycrystalline Materials

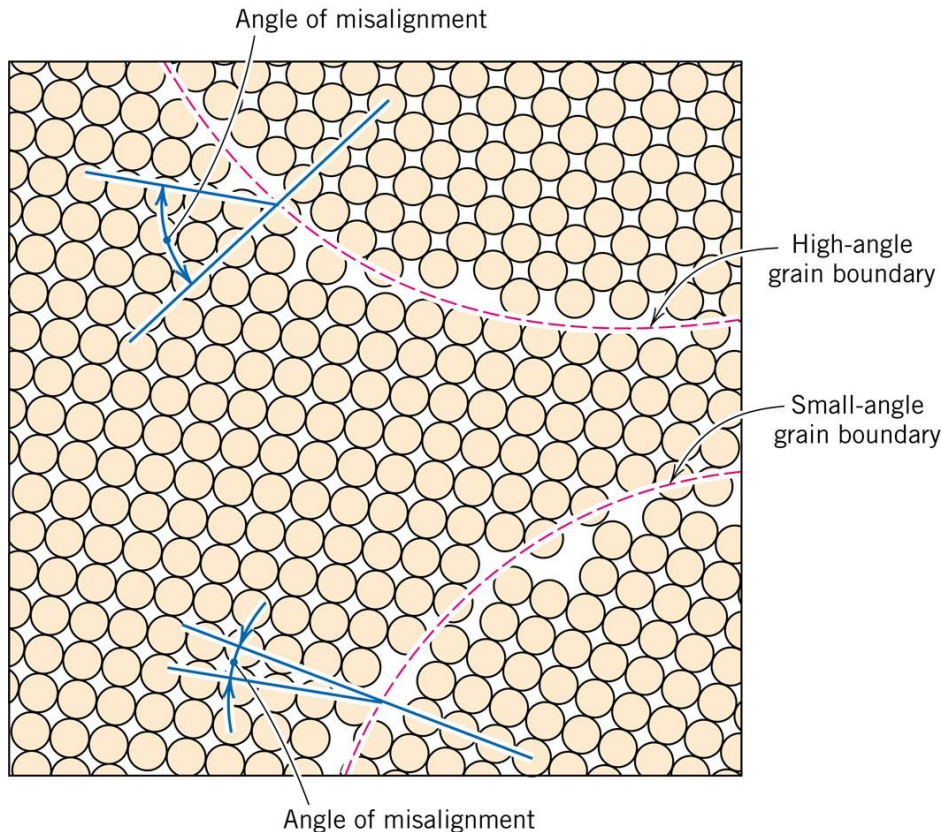
Most materials are polycrystalline and are made of many single crystals



- during *solidification* the crystals nucleate and grow from the liquid in a *random orientation*
- the *grains* impinge on each other when the solidification is complete
- junction of grains are
.....



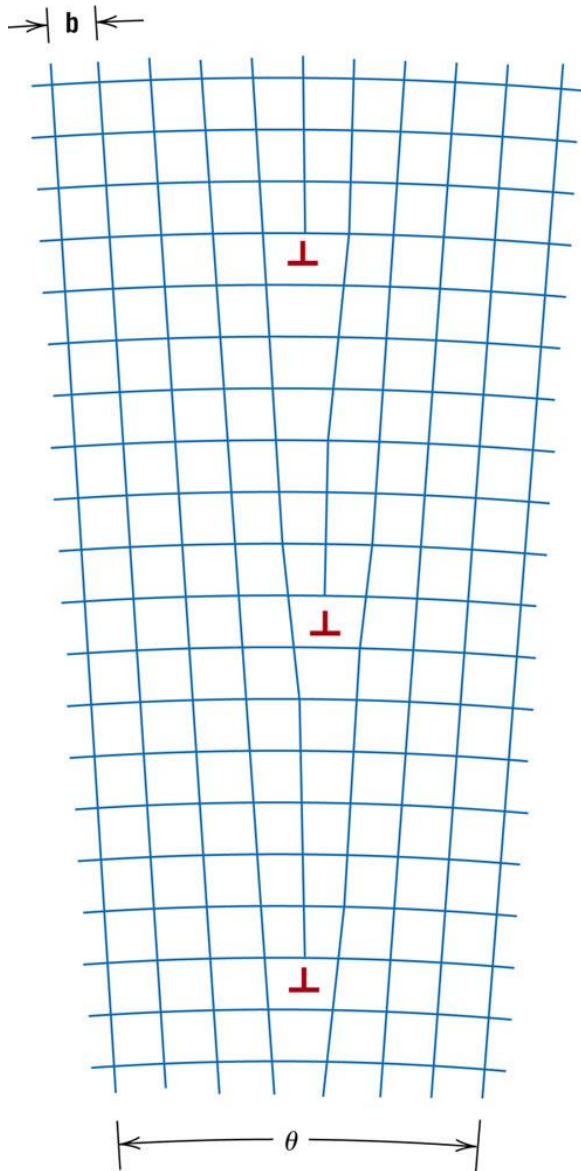
Grain Boundaries



- Occurs due to the crystallographic mismatch when two grains meet
- when mis-orientation is large
→ *high angle grain boundary*
- when mis-orientation is small,
→ *low angle grain boundary*
- atoms are *less bonded* and the atomic packing is lower than in the grain (*lower coordination*)
- the result is an energy difference
→ *interfacial surface energy or grain boundary energy*



Grain Boundaries



- grain boundaries are more chemically reactive
- segregation of impurities due to higher energy
- total grain boundary area smaller in *coarse grained* than *fine grained* material
- low angle grain boundary is described as an array of dislocations
 - *tilt boundary* (edge)
 - *twist boundary* (screw)



Observation of Grain Structure

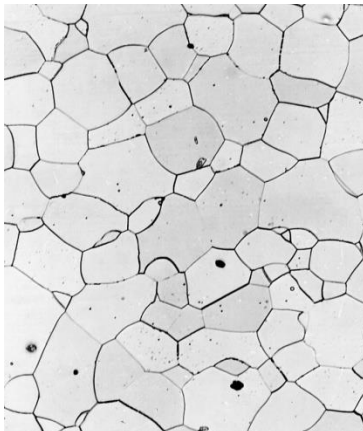
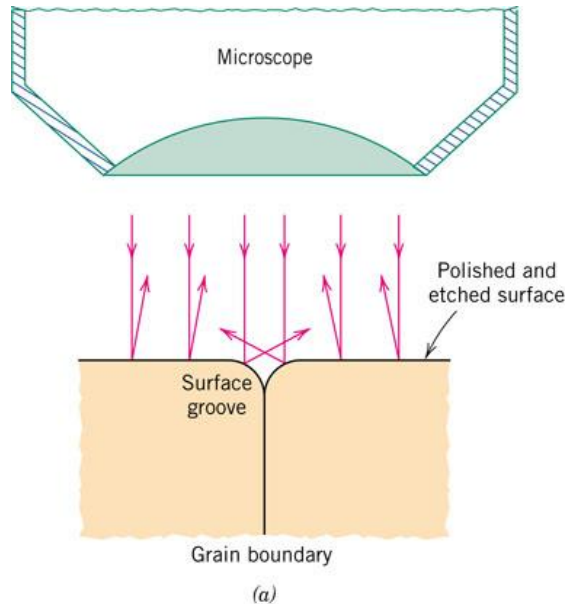


FIGURE 4.10 *High-purity polycrystalline lead ingot in which the individual grains may be discerned.*

- *Macrostructure* can be observed with naked eye
- coarse grains can be revealed this way (e.g. Al streetlight posts, and zinc galvanized garbage cans)
- is when the grains can only be observed with a microscope
- imaged using a camera for archiving
→ *photomicrograph*



Sample Preparation for Microscopy



- Preparation requires meticulous *grinding* and *polishing* of the surface
- the microstructure is revealed by attack using (chemical reagents)
 - preferential attack of grain boundaries
- effect is that these features scatter the incident light and create **optical contrast**



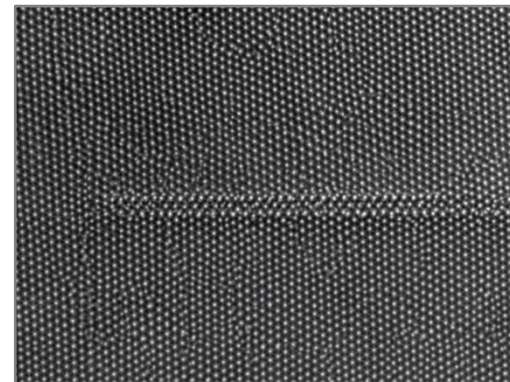
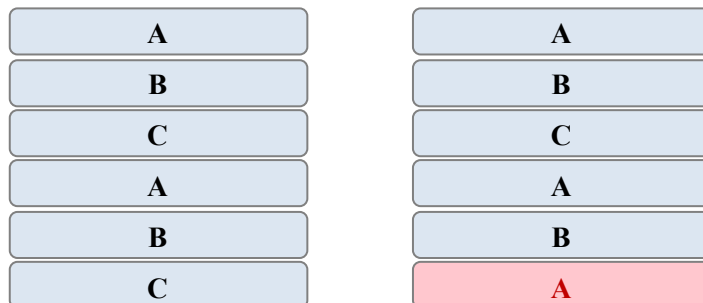
Stacking Faults

- An **area (planar) defect** – a mistake in the stacking sequence of atomic planes.
- Recall: FCC stacks as **ABCABCABC...** and HCP stacks as **ABABAB...**
- A stacking fault is when the order is broken, e.g. **ABCAB**ABC... – one plane (C) is missing.
- The atoms are still in proper positions; only the **order of the planes** is wrong, so a thin HCP-like layer appears inside the FCC crystal.

*Because the mistake is small, its energy is low – stacking faults are common. They form during **solidification and plastic deformation** and affect how easily a metal deforms.*

Compare the stacking sequences:

Perfect FCC: ...ABCABC... Fault: ...ABCAB|A...

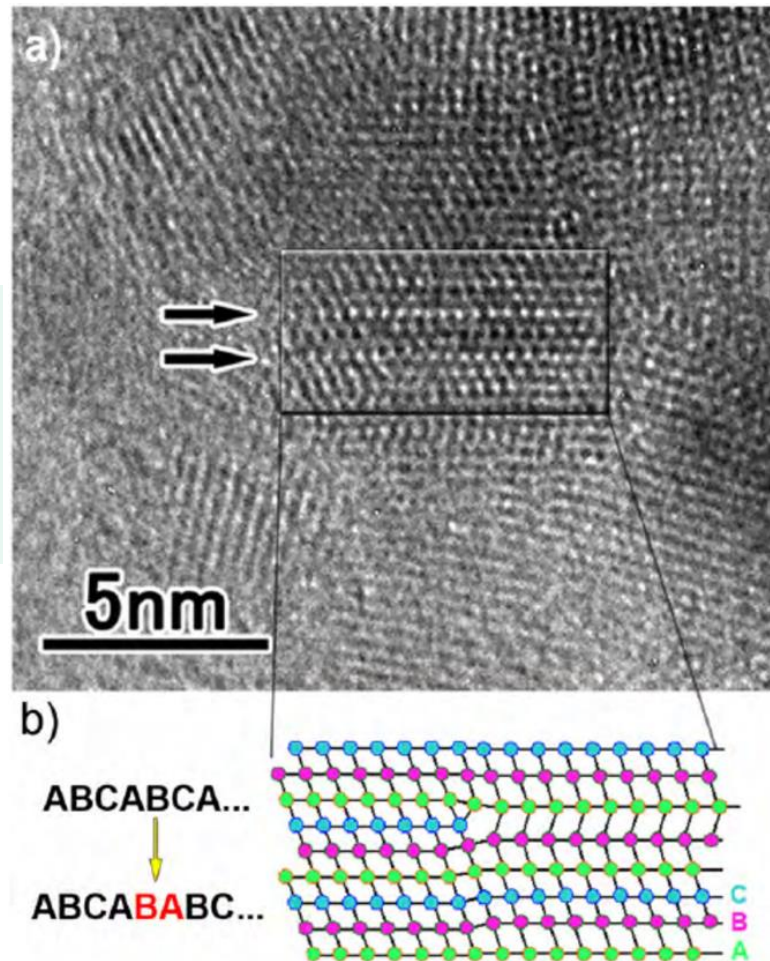


High-resolution TEM: the faulted plane appears as a break in the regular stacking



Stacking Faults

Wurtzite-type ZnS nanoparticles by pulsed electric discharge



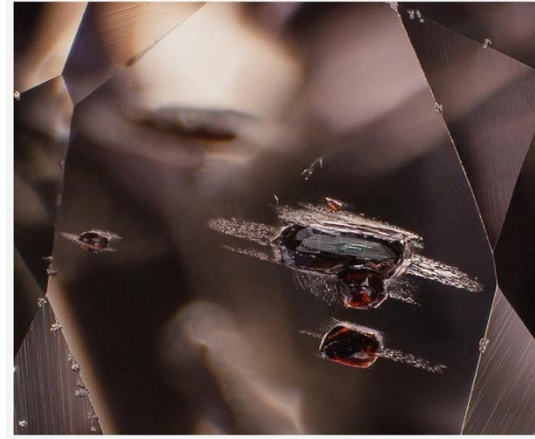
*Omurzak et al. (2011)
Nanotechnology.*

HRTEM image of a stacking fault (a), denoted by the arrows, and (b) schematic illustration of the stacking fault in the crystal shown in (a).

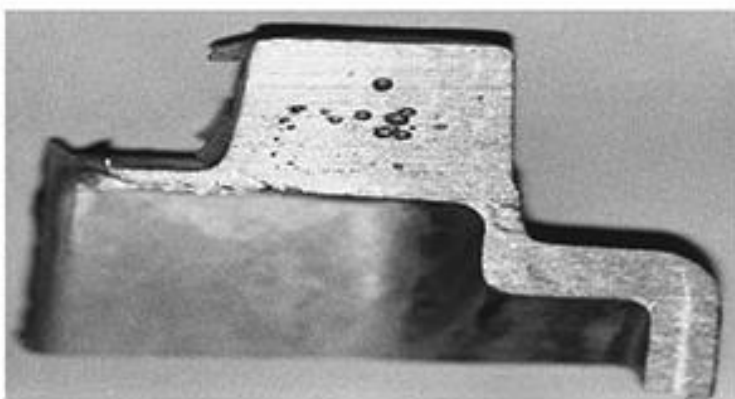


Three-dimensional (*volumetric*) defects

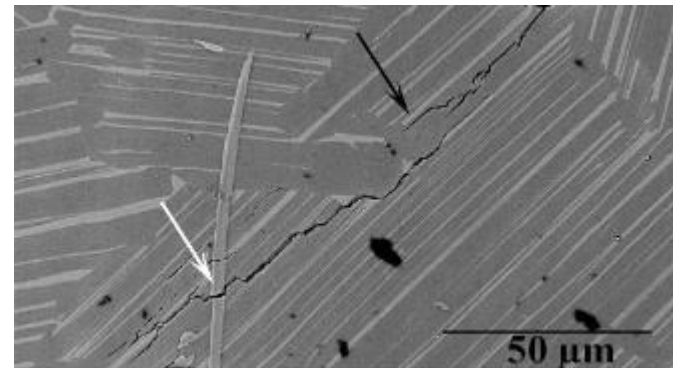
- Inclusions
- precipitates
- Porosities or voids
- Cracks



Sapphire crystal containing Rutile inclusions



Gas porosity in metallic component



Cracks in titanium aluminide

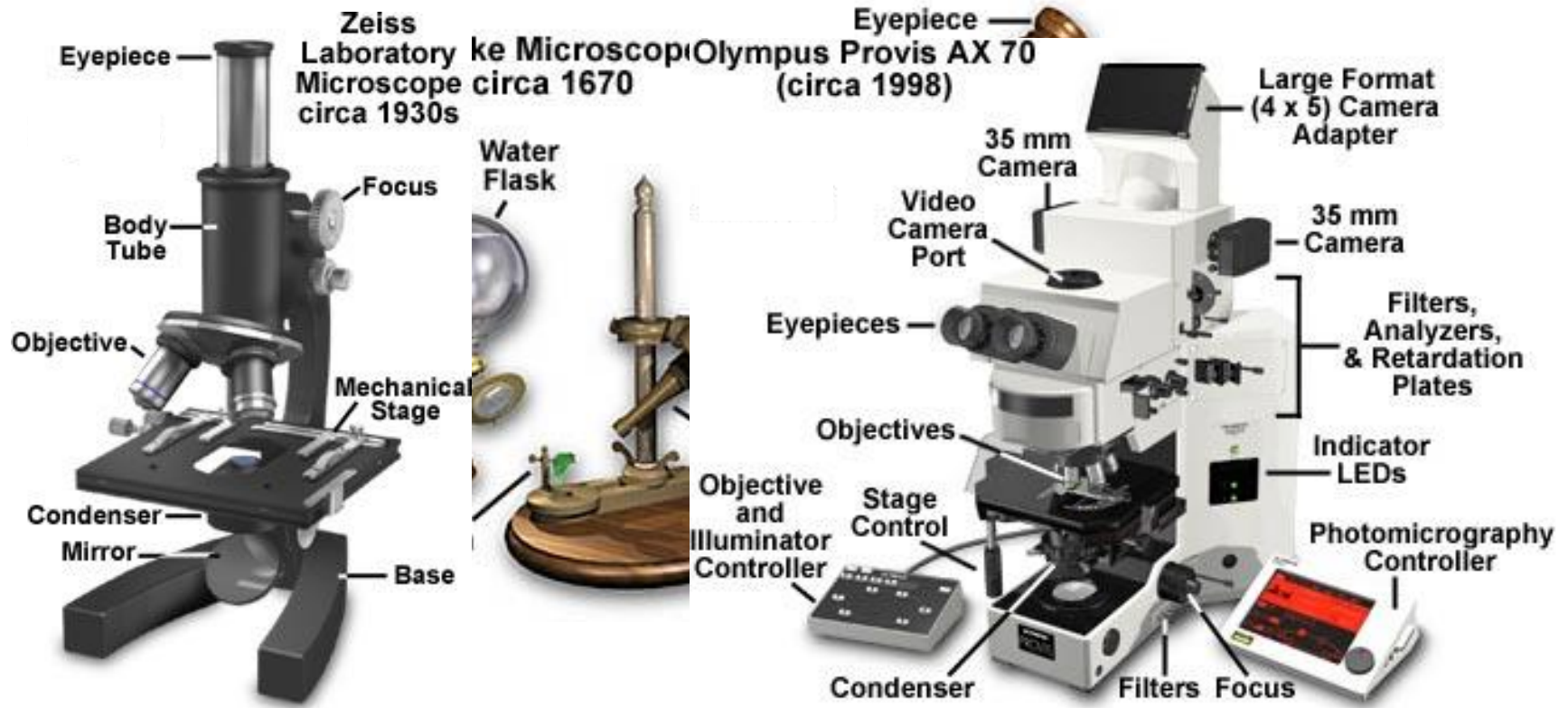


Types of Microscopy

- *Optical microscopy is limited to x2000 magnification*
- *electron microscopy* uses **electron waves** (very small $\lambda \sim 0.003\text{nm}$) rather than light ($\lambda \sim 500\text{nm}$).
- can reveal microstructural features down to atomic scale (x1,000,000)
- scanning electron microscope (.....)
 - sample preparation similar to optical microscopy
 - can be used to observe fracture surface (*fractography*)
- transmission electron microscope (.....)
 - samples are very small
 - requires very thin (***electron transparent***) samples



Optical Microscopy

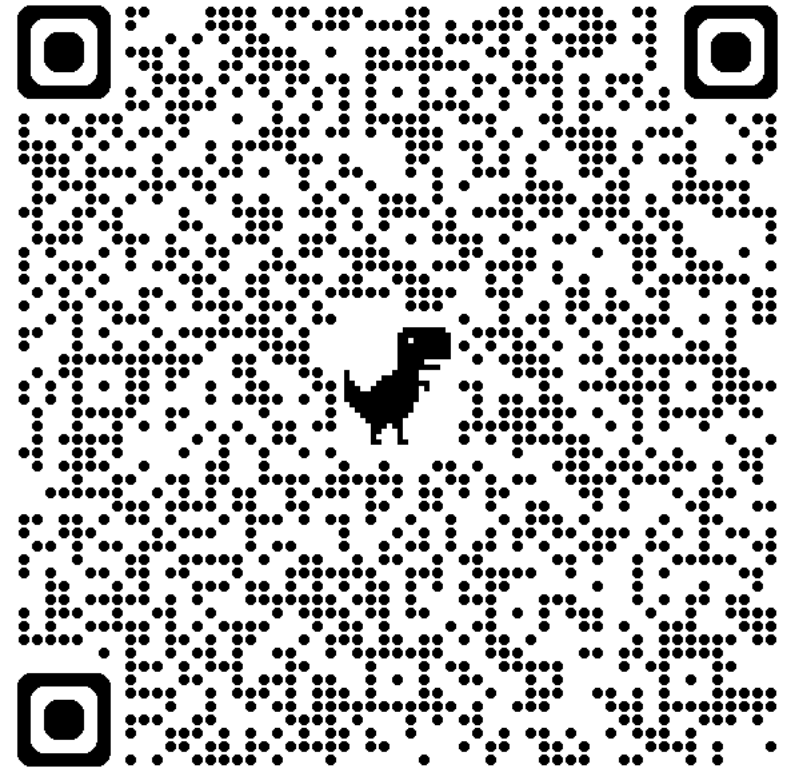




Review Optical Microscopy

A polished metal sample is placed under an optical microscope, but no grains can be seen — the surface just looks like a uniform bright mirror. After the sample is etched, the grains and their boundaries become clearly visible. What is the best explanation?

- A.** Etching increases the magnification so that the grains become large enough to see.
- B.** Etching colours each grain with a different chemical dye, and the optical microscope detects the colours.
- C.** Etching attacks grain boundaries preferentially, so the surface is no longer uniform and these features scatter the incident light differently, creating contrast.
- D.** Before etching the sample was a single crystal.



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Scanning Electron Microscope (SEM)



The SEM is designed for direct studying of the surfaces of solid objects

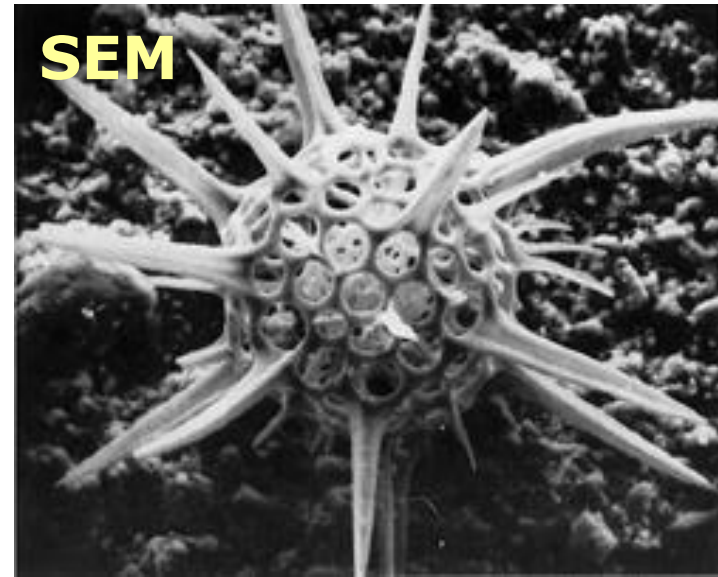
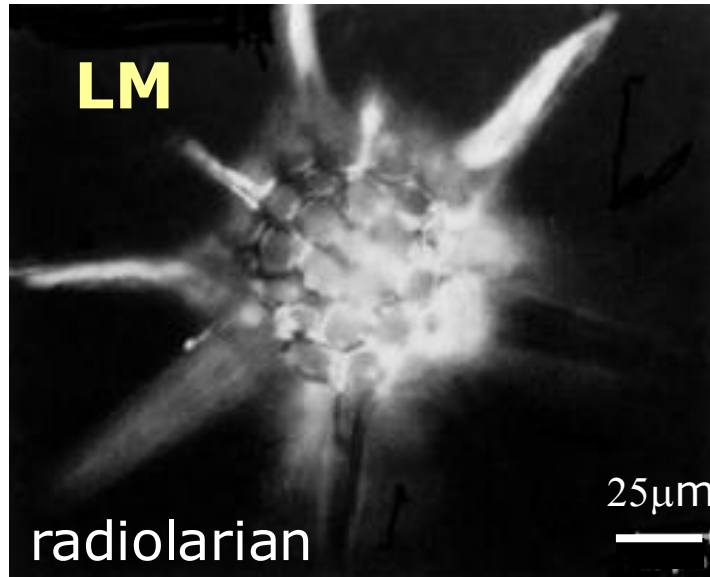
*Scanning electron microscope (SEM) is a microscope that uses **electrons** rather than **light** to form an image. There are many advantages to using the SEM instead of an OM.*



Advantages of Using SEM over LM

- **The SEM has a large depth of field,**

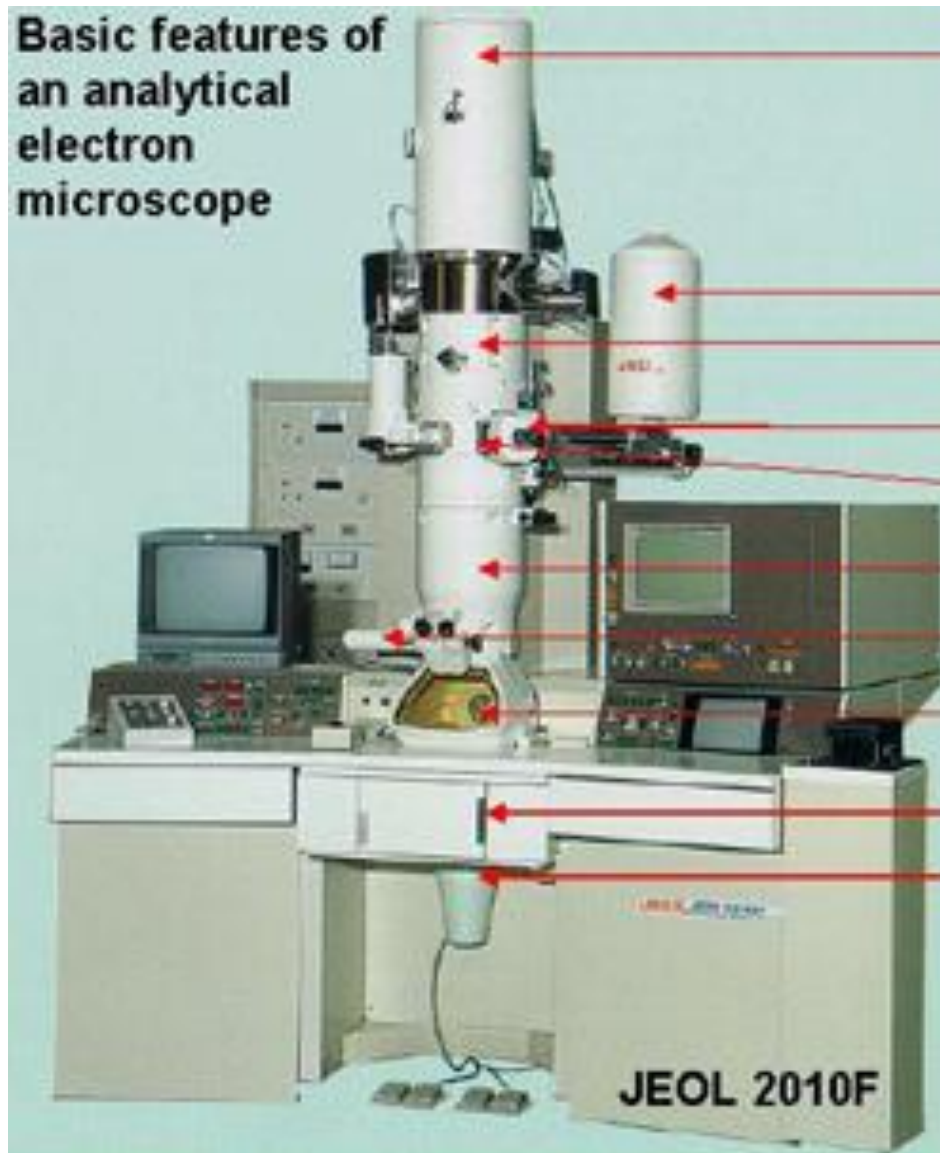
This allows a large amount of the sample to be in focus at one time and produces an image that is a good representation of the three-dimensional sample.



- **The SEM also produces images of high resolution; *fine features can be closely examined at a high magnification.***
- *The combination of higher magnification, larger depth of field, greater resolution makes the SEM one of the most heavily used instruments in research areas and industries, especially in **semiconductor** industry.*



Transition Electron Microscope (TEM)



Basic features of
an analytical
electron
microscope

Electron Gun

EDS Detector
Condenser Lens

Specimen Holder
Objective Lens

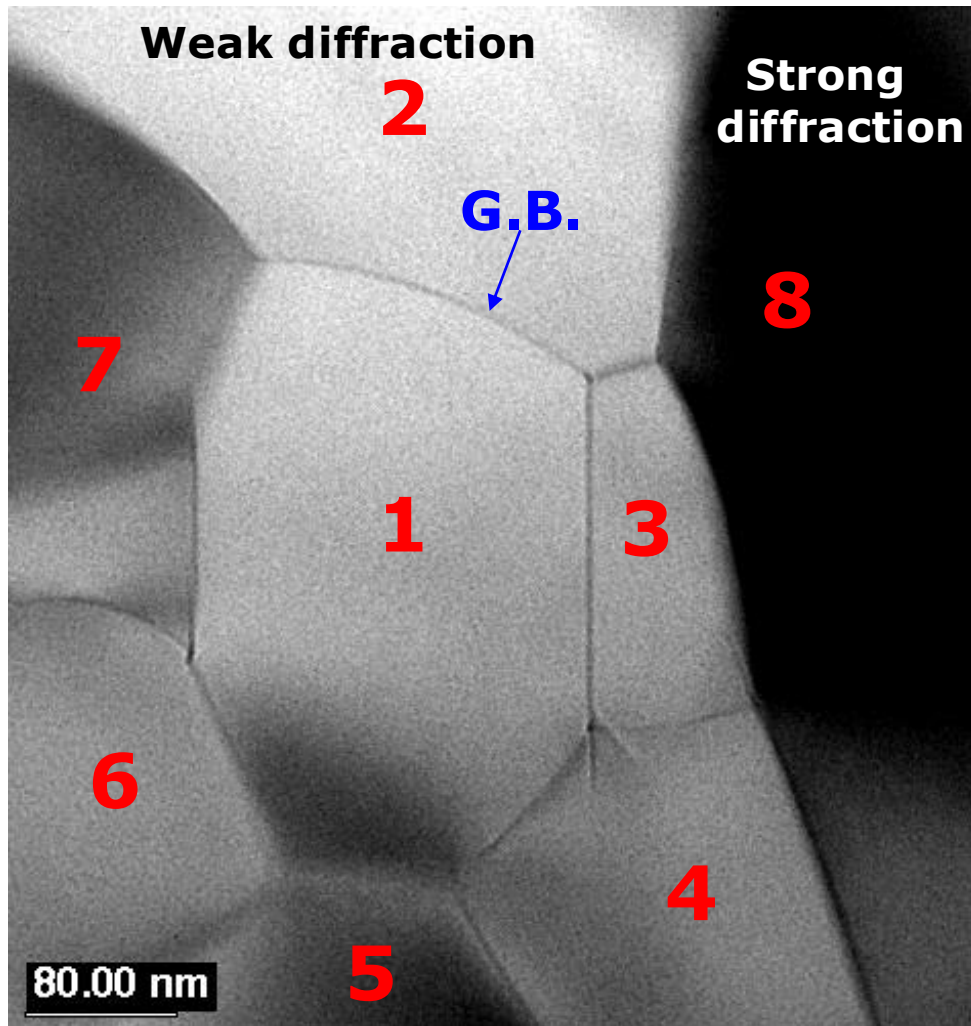
Magnifying Lenses

HAADF Detector
Viewing Chamber

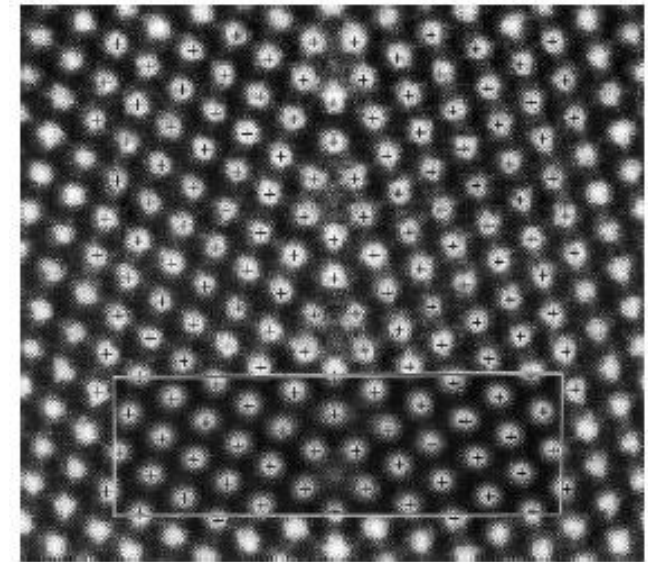
Camera Chamber
**STEM Detector or
EELS**



Transition Electron Microscope (TEM)



8 grains are in different orientations

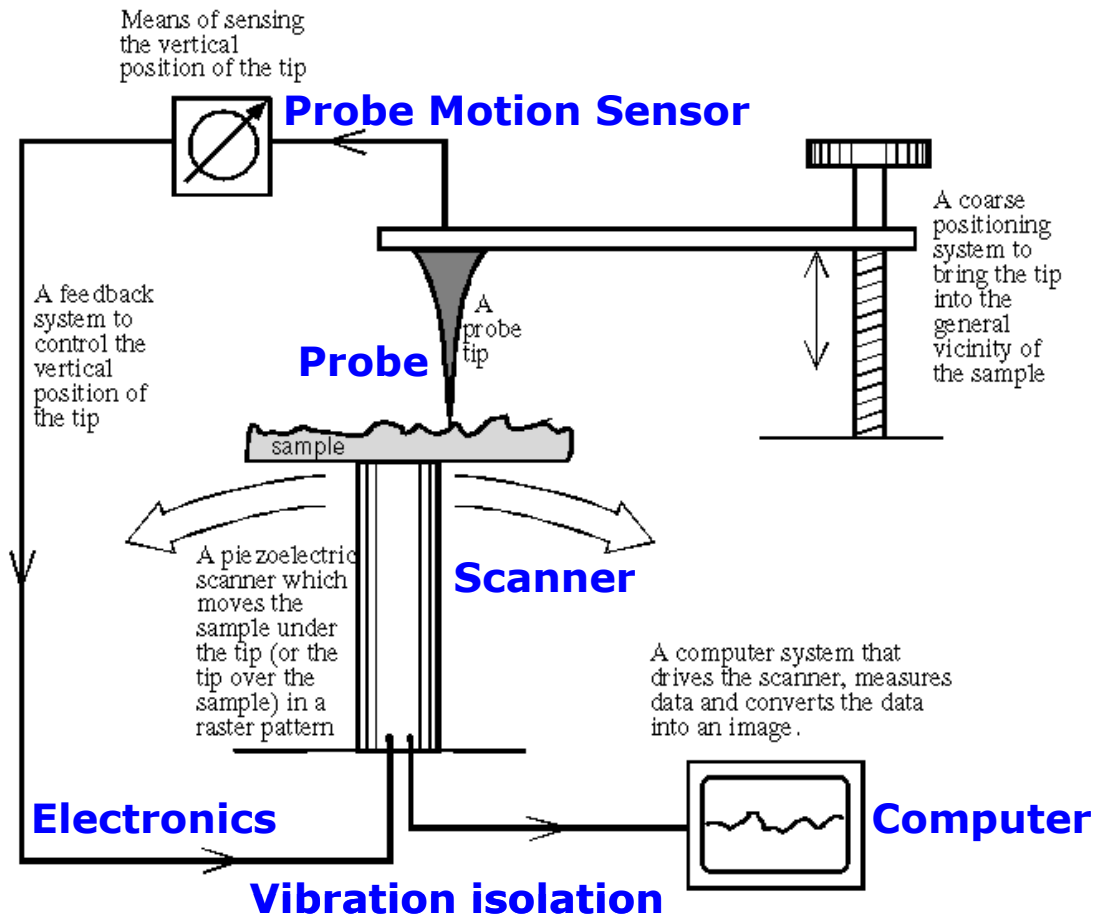


*High-resolution TEM image
of a tilt grain boundary in Al,
Sandia National Lab.*



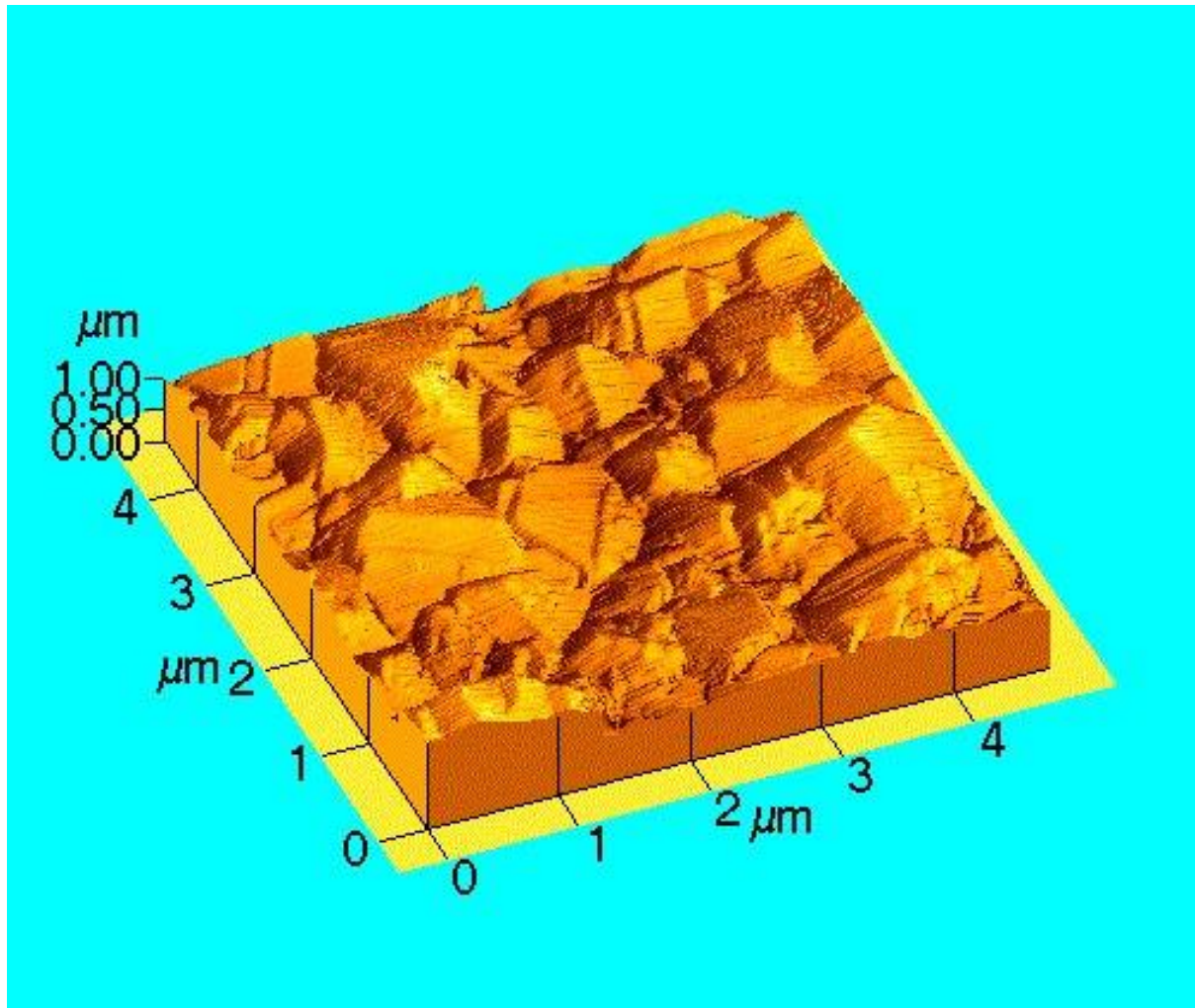
Scanning Probe Microscope (SPM)

SPMs are used for studying surface topography and properties of materials from the atomic to the micron level.





SPM





Microscopy

Question:

Why were commercial TEMs developed from about 1938 and SEMs from about 1965, whereas SPMs were not around before 1980's?

Answer:

TEMs and SEMs were technology initially as images could be easily collected on photographic film which has been around for a long time, with high resolution. On the other hand, SPMs depend on computers to construct the 3D images.



General Resolution of Microscopes

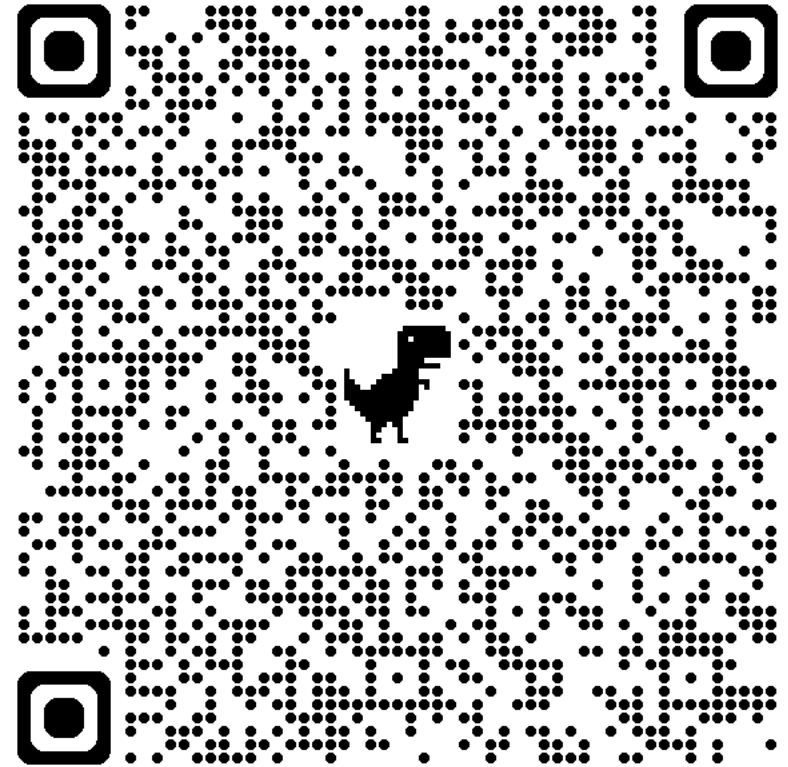
Type of Microscope	Approx. Resolution
Human Eye	 Å
Optical Light (OLM)	3000 Å
Scanning Electron (SEM)	10-50 Å
Transmission Electron (TEM)	2-5 Å, near atomic
Scanning Probe (SPM)	... Å, atomic



Review Electron Microscopy

A fracture surface is very rough, with peaks and valleys. Why does the SEM give a clear image of it while the optical microscope gives an image where only part is in focus?

- A.** The SEM has a much larger depth of field.
- B.** The SEM has higher magnification.
- C.** The SEM uses colour.
- D.** The fracture surface must be etched first.



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Next topic:
Diffusion